

Machine learning surrogates for molecular hydrogen density in three-dimensional interstellar medium simulations: multi-scale spatial features outperform volumetric neural networks under out-of-distribution generalisation

Daniel H. Breger¹★

¹*Department of Physics, Technion, Haifa, Israel*

ABSTRACT

The molecular hydrogen number density n_{H_2} is a fundamental quantity in interstellar-medium physics, yet computing it self-consistently in three-dimensional simulations requires solving time-dependent non-equilibrium chemistry coupled to radiative transfer, an expense that severely limits the resolution and UV-parameter coverage of modern calculations. We train a machine learning surrogate to predict $\log_{10}(n_{\text{H}_2})$ from 15 local physical properties at each cell of a 128^3 simulation grid, and evaluate it under a stringent leave-one- G_0 -out cross-validation protocol in which each fold withholds an entire simulation at a UV field strength G_0 (Habing units) absent from training. Our data set comprises seven UV-only chemistry simulations at $G_0 \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4\}$ Habing, totalling 14,680,064 cells. The principal methodological contribution is the replacement of a 3D U-Net convolutional neural network with multi-scale spatial neighbourhood features: uniform box-filter averages of all 15 physical fields computed at kernel sizes 3^3 , 5^3 , and 7^3 voxels via separable convolution in ≈ 30 s. Adding these 45 spatial features to a gradient-boosted tree model improves the mean log-space R^2 from 0.886 to 0.924 in a single deterministic preprocessing step. An equal-weight ensemble of XGBoost and a wide fully-connected MLP, each trained on the resulting 60-dimensional feature set (`ENS_SP`), achieves a mean R^2 of 0.948 ± 0.024 across all seven folds, with per-fold values ranging from 0.908 at the most demanding extrapolation fold ($G_0 = 0.1$) to 0.967 at mid-range interpolation folds. The 3D U-Net achieves a best mean R^2 of 0.803 ± 0.172 with an order-of-magnitude greater fold-to-fold variance, owing to the mismatch between the volumetric training sample count (48 augmented cubes) and the model parameter count (1.5–23 M). We further demonstrate that dropout regularisation catastrophically degrades extrapolation to unseen G_0 values while L2 weight decay is benign, and that per-fold target normalisation is essential for CNN stability. An intra-cube spatial sampling study shows that 1 per cent uniform random coverage reconstructs the n_{H_2} field with $R^2 = 0.897$, while 50 per cent contiguous-slab coverage yields $R^2 = -0.394$, establishing coverage uniformity as the critical design parameter for observational H_2 density surveys.

Key words: ISM: molecules – ISM: clouds – radiative transfer – astrochemistry – methods: numerical – methods: statistical

1 INTRODUCTION

The formation of molecular hydrogen on dust-grain surfaces (Cazaux & Tielens 2004) and its photodissociation by Lyman–Werner UV photons (Habing 1968; Draine & Bertoldi 1996) together determine the molecular fraction f_{H_2} of the interstellar medium (ISM). Because H_2 is both the dominant gas-phase coolant below ~ 300 K and the direct precursor of CO and other molecular tracers used to infer star-formation rates (Narayanan et al. 2012), f_{H_2} sets the initial conditions for the entire star-formation sequence. Computing f_{H_2} self-consistently within a three-dimensional hydrodynamics simulation requires integrating a stiff system of non-equilibrium chemistry equations at every grid cell, tracking the time-dependent competition between H_2 formation on grain surfaces, UV photodissociation in the Lyman–Werner band (11.2–13.6 eV), and the H_2 self-shielding that occurs when dense H_2 columns attenuate their own dissociating

radiation (Wolcott-Green et al. 2011). In simulations with 10^6 – 10^7 cells, this chemistry solver can dominate the total compute budget; including it at all resolutions and over a wide UV-parameter range is often infeasible (Grassi et al. 2014).

Machine learning (ML) surrogates offer a route to replacing or accelerating the chemistry calculation. A model trained on a finite set of high-fidelity simulations can predict the chemical state in new simulations without running the full solver, reducing the marginal cost of a chemistry-coupled simulation from hours to seconds. This approach has been applied across several regimes of astrophysical chemistry, from primordial gas cooling in early-universe simulations to CO chemistry in turbulent molecular clouds (Glover & Mac Low 2007; Glover et al. 2010), and has attracted growing interest as the resolution and scope of ISM simulations increases (Bisbas et al. 2021; Katz et al. 2022).

The shared challenge in these settings is threefold. First, the target quantity n_{H_2} spans approximately 35 orders of magnitude in linear space between UV-exposed ($n_{\text{H}_2} \approx 10^{-30} \text{ cm}^{-3}$) and deeply shielded

★ E-mail: danielbreger@campus.technion.ac.il

($n_{\text{H}_2} \approx 10^5 \text{ cm}^{-3}$) gas, making regression in linear space numerically intractable. Second, the transition between phases is sharp and exponentially sensitive to temperature: the H^0/H_2 boundary occurs over less than a factor of two in column density. Third, a cell's chemical state is not fully determined by its local physical properties alone. Whether a cell is shielded from dissociating UV depends on the column density of gas and dust between it and the UV source — an inherently non-local quantity. Per-cell models must rely on proxy quantities such as the self-shielding factor or local extinction to recover this spatial information.

A volumetric convolutional neural network (CNN) naturally addresses the non-locality problem: by processing the entire simulation volume, it can in principle learn to aggregate spatial context through its receptive field. However, CNN-based surrogates for ISM chemistry have received limited systematic evaluation. When the available training data consists of only a handful of simulation cubes at discrete parameter values, the mismatch between volumetric sample count and model parameter count can be severe, and the stability of CNN predictions on out-of-distribution parameter values is unclear.

In this paper we address these questions systematically on the UV-only chemistry suite, seven 128^3 -cell simulations spanning a 64-fold range in UV field strength G_0 . We compare a hierarchy of models ranging from linear regression through gradient-boosted trees, fully-connected MLPs, and a 3D U-Net, evaluating all under the demanding protocol of leave-one- G_0 -out cross-validation. Our principal finding is that *engineered spatial features* — precomputed multi-scale box-filter averages of local physical fields — provide spatial context more reliably and efficiently than learned volumetric representations, achieving both higher accuracy and lower fold-to-fold variance than the 3D CNN at a fraction of the computational cost.

The paper is structured as follows. Section 2 describes the simulation dataset. Section 3 defines the evaluation protocol. Section 4 describes feature engineering and model architectures. Results are presented in Section 5, supplementary experiments in Section 6, and the discussion and conclusions in Sections 7 and 8.

2 DATA

2.1 Simulation suite

The data set consists of seven three-dimensional uniform-grid simulations from the UV-only chemistry suite. Each simulation occupies a $128 \times 128 \times 128$ grid of 2,097,152 cells, evolving identical initial conditions of turbulent ISM gas under a spatially uniform, time-steady UV radiation field of strength G_0 measured in Habing units (Habing 1968). The seven simulations span a 64-fold dynamic range: $G_0 \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4\}$ Habing, sampled on a geometric sequence with a factor-of-two spacing. Adjacent simulations are physically similar enough for cross-validation to be meaningful, while the full 64-fold range produces qualitatively distinct chemical equilibria. The total data set contains 14,680,064 cells.

2.2 Physical fields and feature construction

Each cell carries 18 raw fields. Three coordinate fields (i_x, i_y, i_z) serve as grid indices and are excluded from model input. The target is n_{H_2} (cm^{-3}). The remaining 15 fields constitute the input feature set.

Three fields spanning many orders of magnitude — total hydrogen number density n_{H} , gas temperature T , and ionised hydrogen density

n_{H^+} — are log-transformed:

$$x_\ell = \log_{10}(x + \varepsilon), \quad \varepsilon = 10^{-30}, \quad (1)$$

where ε prevents singularities without affecting physics at any physically realised value. The H_2 self-shielding factor f_{H_2} is similarly log-transformed and merits special attention: it measures the fraction of Lyman–Werner photons attenuated by the line-of-sight H_2 column between each cell and the UV source, computed by the chemistry solver as a non-local integral along the illumination axis. Two cells with identical local n_{H_2} can have very different f_{H_2} values depending on their position within the cloud; f_{H_2} is therefore not algebraically derivable from local properties and constitutes a legitimate, highly informative input feature. The remaining fields are the dust extinction proxy ext , the global UV field strength $\log G_0$ (constant within each cube), the three gas velocity components (v_x, v_y, v_z) tracing turbulent motions, and six face-centred magnetic field components ($B_{xl}, B_{xr}, B_{yl}, B_{yr}, B_{zl}, B_{zr}$).

2.3 Target distribution

Figure 1 shows the distribution of $\log_{10}(n_{\text{H}_2})$ across six representative G_0 values. At $G_0 = 0.1$ the distribution is bimodal, with a UV-exposed population near $\log_{10}(n_{\text{H}_2}) \approx -10$ and a self-shielded molecular population near $\log_{10}(n_{\text{H}_2}) \approx 0$. As G_0 increases, photodissociation pushes the shielded peak to lower densities and eventually merges the two populations into a single UV-dominated mode. The full dynamic range spans ~ 35 dex in linear space, making regression in that space numerically intractable. In $\log_{10}$ space the distribution is approximately Gaussian, which is why log-space prediction is both numerically necessary and statistically natural.

3 EVALUATION PROTOCOL

3.1 Leave-one- G_0 -out cross-validation

We adopt leave-one- G_0 -out seven-fold cross-validation: each fold withholds one entire simulation cube (≈ 2.1 M cells at a single G_0) and trains on the remaining six (≈ 12.6 M cells). This design was chosen over random k -fold splitting because random splits would allow the model to see cells from every G_0 value during training. Since all cells within a cube share the same global UV intensity, a randomly split model faces only within-distribution interpolation, grossly inflating apparent performance. The real-world application is to predict n_{H_2} for a simulation at a UV field strength not yet computed; leave-one- G_0 -out directly measures this out-of-distribution capability.

The difficulty of each fold depends on whether the held-out G_0 falls within the convex hull of training values (interpolation) or outside it (extrapolation). The two boundary folds — $G_0 = 0.1$, for which no training data lies below, and $G_0 = 6.4$, above which no training data lies — require downward and upward extrapolation respectively. All interior folds require interpolation between a pair of bracketing training cubes. This a-priori prediction is confirmed by every experiment: $G_0 = 0.1$ and $G_0 = 0.2$ are consistently the hardest folds. The upper boundary fold $G_0 = 6.4$ is intermediate in difficulty because at high UV intensity photodissociation dominates uniformly, producing a simpler chemistry than the self-shielding dominated regime at low G_0 .

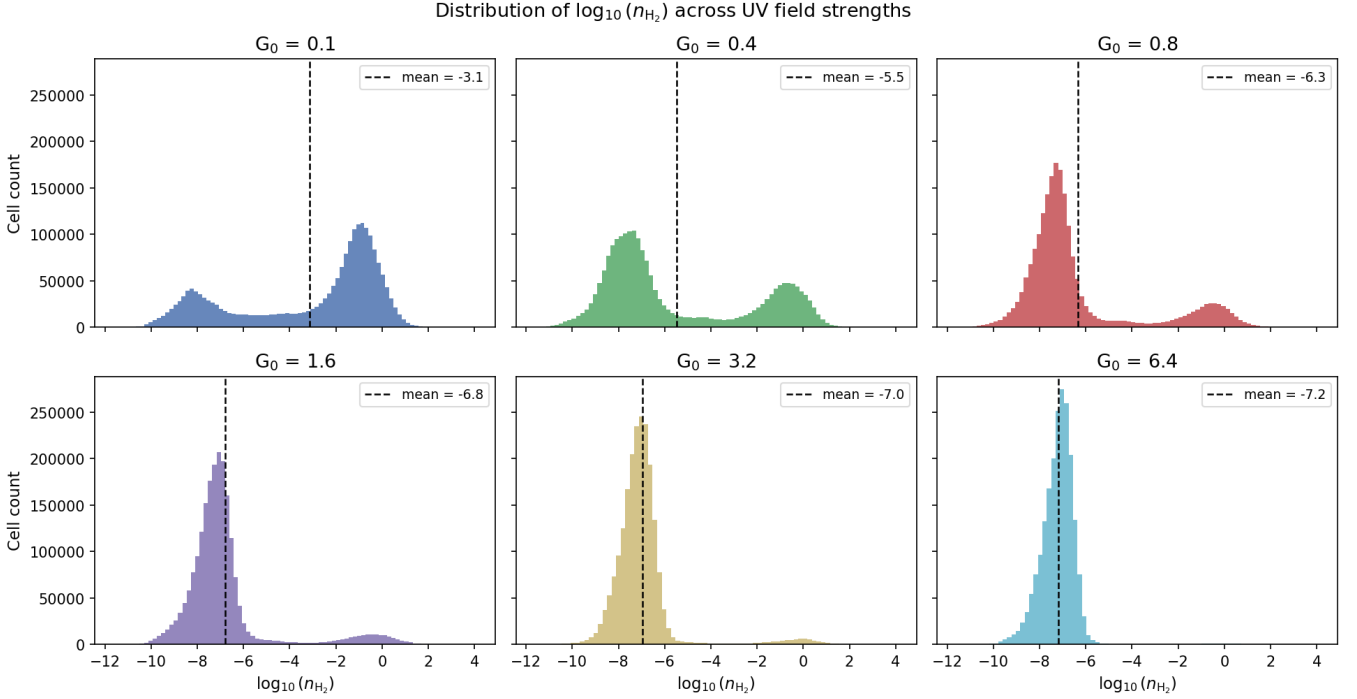


Figure 1. Distribution of $\log_{10}(n_{H_2})$ across six UV field strengths (the seventh cube, $G_0 = 0.2$, is omitted for clarity). Each panel shows the cell-count histogram for one simulation cube. The bimodal structure visible at $G_0 = 0.1$ (UV-exposed and self-shielded populations) progressively collapses to a single UV-dominated mode as G_0 increases. Dashed vertical lines mark per-cube means, which shift from -3.1 at $G_0 = 0.1$ to -7.2 at $G_0 = 6.4$ as H_2 photodissociation becomes increasingly dominant. All modelling is performed in log-space.

3.2 Metrics

All primary results are reported in $\log_{10}(n_{H_2})$ space (dex). The coefficient of determination

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (2)$$

is the primary metric. $R^2 = 0.95$ corresponds to typical per-cell errors of ~ 0.1 – 0.2 dex, equivalent to a factor of 1.3–1.6 in linear n_{H_2} . The root mean square error (RMSE) and mean absolute error (MAE) in dex quantify the magnitude and distribution of errors respectively. As a secondary metric, a clipped linear-space R^2 is computed by exponentiating predictions clipped to ± 1 dex of the true range before computing equation (2), providing a bounded linear-space comparison.

4 METHODS

4.1 Multi-scale spatial neighbourhood features

The dominant non-local effect governing n_{H_2} is the attenuation of Lyman–Werner photons by surrounding gas. A cell’s effective UV exposure depends on the column density of gas and dust between it and the UV source, information encoded in the 3D spatial arrangement of neighbours. To provide pointwise models with spatial context without the complexity and instability of a full volumetric CNN, we precompute multi-scale box-filter averages of all 15 physical features.

For each feature f and simulation cube, we reshape the $N = 2,097,152$ tabular values into a 128^3 volume and apply a uniform

box filter at kernel half-widths $k \in \{3, 5, 7\}$:

$$\bar{f}_{i,k} = \frac{1}{k^3} \sum_{j \in \mathcal{N}_k(i)} f_j, \quad (3)$$

where $\mathcal{N}_k(i)$ denotes the k^3 voxels centred on grid position i . Because the box filter is separable, the cost is $\mathcal{O}(N)$ per scale regardless of k . The total compute time for all 15 features at all three scales across all seven cubes is approximately 30 seconds. The filtered values are re-indexed to tabular form using the original grid coordinates, yielding $15 \times 3 = 45$ spatial features per cell. These are concatenated with the 15 baseline features to produce a 60-dimensional representation:

$$\mathbf{x}_i^{(60)} = \left[\mathbf{x}_i^{(15)} \parallel \bar{\mathbf{x}}_{i,3}^{(15)} \parallel \bar{\mathbf{x}}_{i,5}^{(15)} \parallel \bar{\mathbf{x}}_{i,7}^{(15)} \right]. \quad (4)$$

Figure 2 illustrates the contrast between traditional per-cell modelling and this approach. The three scales encode complementary physical information: the 3^3 kernel captures local density and temperature gradients; the 5^3 kernel captures the cloud environment over which self-shielding columns become significant; and the 7^3 kernel captures meso-scale structure such as the cloud–inter-cloud boundary.

4.2 Model architectures

4.2.1 Linear regression

Ordinary least-squares regression with feature standardisation serves as a lower bound, testing whether the n_{H_2} relationship is approximately linear after log-transformation.

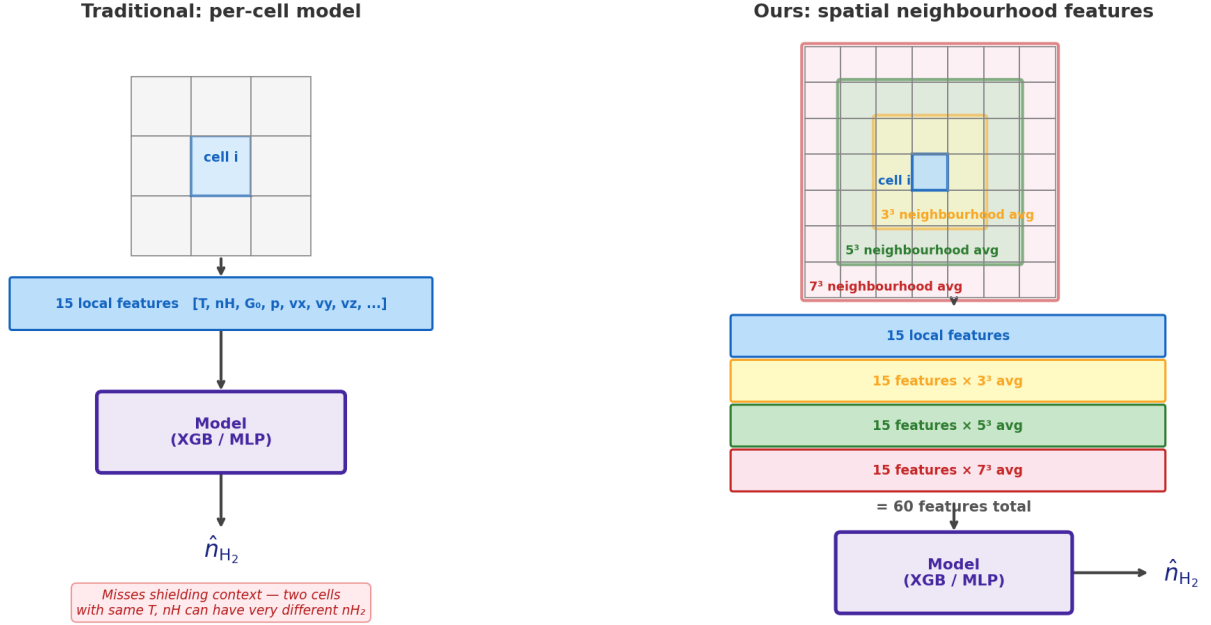


Figure 2. Schematic comparison of the traditional per-cell modelling approach (*left*) and the multi-scale spatial neighbourhood feature approach introduced in this work (*right*). A traditional model receives only the 15 local physical properties at cell i , missing the shielding context that distinguishes two cells with identical (T, n_H) but very different n_{H_2} . Our approach augments the local feature vector with box-filter averages of all 15 features at three nested kernel sizes ($3^3, 5^3, 7^3$ voxels), producing a 60-dimensional feature vector that encodes the spatial environment of each cell without requiring volumetric model training.

4.2.2 Gradient-boosted trees

We use XGBoost (Chen & Guestrin 2016) with the GPU-accelerated histogram tree method. Each fold trains on ≈ 12.6 M cells with 400 boosting rounds, maximum tree depth 6, learning rate 0.1, and a row subsampling fraction of 0.3. Three depth variants (4, 6, 8) were evaluated; all produced $R^2 \approx 0.886$ (differences < 0.002), confirming that the primary predictive signal comes from low-order feature interactions and that tree depth is not a bottleneck.

4.2.3 Fully-connected MLP

The MLP is implemented in PyTorch and trained on GPU. All training data are preloaded to device memory in a single transfer; batching within each epoch is executed via GPU-native random permutation, eliminating per-batch host-device transfer overhead. Training uses the Adam optimiser (Kingma & Ba 2015) with learning rate 10^{-3} , L2 weight decay 10^{-5} , batch size 262 144, 100 epochs per fold, and a cosine annealing learning rate schedule. Three architectures were evaluated: MLP_STANDARD with widths [256, 256, 128, 64] and batch normalisation (Ioffe & Szegedy 2015); MLP_WIDE with [512, 512, 256, 128]; and MLP_RESIDUAL with four residual blocks (He et al. 2016) of width 256. All three produced $R^2 \approx 0.885$ (differences < 0.003), confirming that model capacity is not the bottleneck; we report MLP_WIDE as the representative result.

4.2.4 3D U-Net

The volumetric CNN is a 3D U-Net (Ronneberger et al. 2015) with three encoder downsampling stages ($64^3 \rightarrow 32^3 \rightarrow 16^3$) via $3 \times 3 \times 3$ convolutions with max-pooling, a bottleneck at 8^3 spanning the full input volume, and a symmetric decoder with trilinear upsampling and skip connections. Because the batch size is necessarily 1 (one

entire 64^3 volume), batch normalisation is replaced with instance normalisation (Ulyanov et al. 2017), which normalises spatial dimensions per channel and is stable at any batch size. Input volumes are downsampled from 128^3 to 64^3 by average pooling before entering the network, reducing memory requirements by $8\times$. Three capacity variants are compared: UNET_SMALL (16 base channels, 1.5 M parameters), UNET_STANDARD (32 base channels, 5.8 M parameters), and UNET_LARGE (64 base channels, 23 M parameters).

4.2.5 Physics-aware data augmentation for the CNN

The cubic simulation grid has the full O_h octahedral symmetry group of 48 operations. However, the UV field illuminates the cube along the $+z$ axis, breaking any operation that interchanges z with a transverse axis. The physically valid augmentations are the eight z -preserving operations (four rotations about z and four reflections through planes containing z). Scalar fields (density, temperature, etc.) transform by index permutation. Polar vector fields (velocity) transform as $\mathbf{v}' = \mathbf{R}\mathbf{v}$. Axial vector fields (magnetic field) transform as $\mathbf{B}' = \det(\mathbf{R})\mathbf{R}\mathbf{B}$, with additional left-right face index swaps on axes whose sign is negated by $\mathbf{R}$. Using the full 48-element group injects physically invalid training samples and was verified to degrade results.

4.3 Ensemble strategy

The primary ensemble (ENS_SP) combines the per-cell predictions of XGB_STANDARD_SP and MLP_WIDE_SP (both trained on the 60-feature set) with equal weights:

$$\hat{y}_i^{\text{ENS_SP}} = \frac{1}{2} \hat{y}_i^{\text{XGB}} + \frac{1}{2} \hat{y}_i^{\text{MLP}}. \quad (5)$$

XGBoost produces piecewise-constant predictions defined by axis-aligned decision boundaries and extrapolates conservatively (con-

Table 1. Incremental ablation of the ENS_SP model. All R^2 values are computed on $\log_{10}(n_{H_2})$ (dex) under leave-one- G_0 -out cross-validation. The delta column gives the improvement over the previous step.

Configuration	Features	Mean R^2	σ	Δ
XGBoost, baseline	15	0.886	0.061	—
+ spatial 3^3	30	0.917	0.032	+0.031
+ spatial $5^3, 7^3$	60	0.924	0.026	+0.007
+ MLP_sp (equal-weight)	60	0.948	0.024	+0.024

stant leaf values) beyond the training feature range. The MLP produces smooth, continuous predictions accurate in regions of smooth feature variation. These complementary inductive biases produce correlated but non-identical errors, and their average is consistently superior to either component alone (Section 5.3).

A Ridge-regression meta-learner (STACKED_SP) that learns fold-specific combination weights was also evaluated. It is trained on out-of-fold predictions from the six meta-training folds for each target fold.

4.4 Regularisation

A central and reproducible finding of this work is that dropout (Srivastava et al. 2014) is harmful in this setting. Adding dropout of probability 0.10 to the CNN encoder and 0.20 to the bottleneck caused the $G_0 = 6.4$ extrapolation fold to collapse from $R^2 = +0.22$ to $R^2 = -0.61$, a catastrophic failure. With only six distinct G_0 values in training, the model must form coherent representations of how n_{H_2} responds to changes in G_0 and its interaction with local conditions. Dropout randomly zeros feature channels, forcing redundant information encoding and preventing the structured representations needed for coherent extrapolation beyond the training range. For interpolation folds this degradation is modest; for extrapolation folds it is catastrophic. An equivalent instability was observed with DART (dropout additive regression trees) applied to XGBoost. L2 weight decay ($\lambda = 10^{-5}$) constrains parameter magnitudes without disrupting their structural organisation and is used as the sole regulariser throughout.

5 RESULTS

5.1 Ablation study

Table 1 presents the incremental gains from the spatial features and ensemble, and Table 2 the complete per-model, per-fold comparison. Figure 3 visualises the ablation with individual fold dots.

The largest single-step improvement (+0.031) is the 3^3 spatial features, which exceeds the combined gain from extending to three scales (+0.007) and adding the MLP ensemble (+0.024). Neural architecture search over three qualitatively distinct MLP designs and tree depth search over the range 4–8 produced differences smaller than 0.003, confirming that feature engineering is the primary bottleneck, not model capacity.

5.2 Per-fold analysis and error characterisation

Figure 4 provides a three-panel summary of the final model analysis. The per-fold R^2 values for ENS_SP (panel a) follow the a-priori difficulty ordering: $G_0 = 0.1$ (0.908) and $G_0 = 0.2$ (0.915) are the hardest; interior folds ($G_0 = 0.4$ – 3.2) all exceed 0.95; $G_0 = 6.4$ achieves 0.965. The relative ease of the upper boundary reflects a physical asymmetry: at high G_0 , UV photodissociation dominates

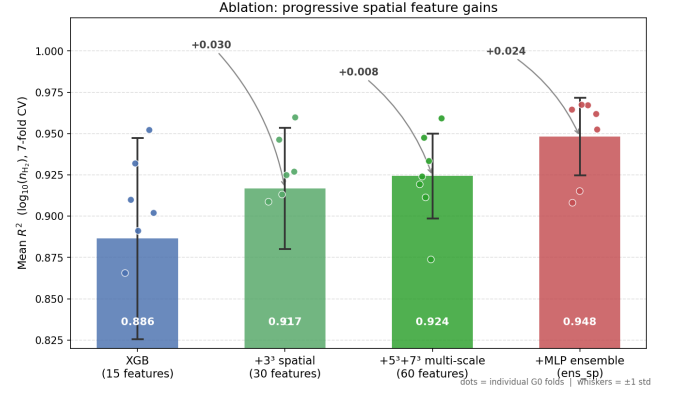


Figure 3. Progressive improvement in mean leave-one- G_0 -out R^2 at each step of the ablation. Bars show the mean; whiskers indicate $\pm 1\sigma$ across folds; filled and open dots show individual fold R^2 values (open dots: extrapolation folds $G_0 = 0.1$ and $G_0 = 0.2$). The 3^3 spatial step contributes the largest single-step gain (+0.030), exceeding the combined effect of all subsequent steps.

uniformly and the equilibrium n_{H_2} is set by a simpler density–UV balance, whereas the low- G_0 regime requires the model to correctly represent the self-shielding runaway that creates the high-density molecular phase. The fold-to-fold standard deviation ($\sigma = 0.024$) is an order of magnitude smaller than that of the best 3D U-Net ($\sigma = 0.172$).

Figure 5 shows predicted versus true $\log_{10}(n_{H_2})$ for all seven held-out folds. The characteristic bimodal structure of the target distribution is reproduced in all panels: the model correctly separates UV-exposed gas (lower-left cluster) from shielded molecular gas (upper-right cluster) in every fold. Scatter is visibly elevated at $G_0 = 0.1$ and $G_0 = 0.2$, consistent with the quantitative R^2 values, and is tightest at the mid-range interpolation folds.

Figure 6 shows the per-fold residual distributions (predicted – true, in dex). All seven distributions are approximately Gaussian and centred near zero, confirming low bias. The distribution width narrows monotonically from the boundary folds to the interior folds. A mild systematic positive bias at $G_0 = 0.1$ indicates that the model slightly over-predicts n_{H_2} in the shielded regime under downward extrapolation, consistent with the expectation that the self-shielding runaway at very low G_0 is harder to extrapolate below the training range.

Figure 7 reveals the density dependence of the remaining errors. Panel (a) shows that restricting evaluation to progressively denser subsets universally improves R^2 , with performance approaching unity for the densest 10 per cent of cells in all folds. Panel (c) shows that errors peak in the second-lowest density decile, at the sharp H/H₂ transition between fully dissociated and partially shielded gas. Panel (d) confirms that even the densest 1 per cent of cells — the most deeply shielded molecular cores — are predicted with $R^2 > 0.9$ in all but the hardest folds.

Figure 8 overlays predicted and true marginal distributions of $\log_{10}(n_{H_2})$ per fold. The ensemble reproduces the multi-modal structure in all folds, placing the UV-exposed and shielded populations correctly. Minor discrepancies in the height of the shielded peak at $G_0 = 0.1$ and $G_0 = 0.2$ are consistent with the positive bias in the residuals.

Table 2. Leave-one- G_0 -out performance (log-space R^2) for all evaluated configurations. The best result in each column is bold. The 3D U-Net results represent the best-performing size variant (UNET_STANDARD, 5.8 M parameters, 150 epochs, no dropout) for each fold where a result was obtained; missing entries indicate runs that were not completed for that fold variant. The linear regression results are shown for context; negative R^2 values indicate performance worse than predicting the fold mean.

Model	Feat.	Mean R^2	σ	$G_0 = 0.1$	$G_0 = 0.2$	$G_0 = 0.4$	$G_0 = 0.8$	$G_0 = 1.6$	$G_0 = 3.2$	$G_0 = 6.4$
ENS_SP (XGB _{sp} + MLP _{sp})	60	0.948	0.024	0.908	0.915	0.953	0.962	0.967	0.967	0.965
STACKED_SP	60	0.946	0.020	0.912	0.916	0.954	0.960	0.962	0.961	0.954
XGB_STANDARD_SP	60	0.924	0.026	0.912	0.874	0.920	0.934	0.924	0.948	0.959
MLP_WIDE_SP	60	0.913	0.034	0.843	0.886	0.914	0.928	0.946	0.945	0.927
ENS_XGB+MLP	15	0.923	0.045	0.891	0.829	0.916	0.949	0.959	0.959	0.954
XGB_STANDARD	15	0.886	0.061	0.891	0.751	0.866	0.902	0.910	0.932	0.952
MLP_STANDARD	15	0.886	0.046	0.809	0.825	0.883	0.921	0.929	0.928	0.910
3D U-Net (STANDARD)	vol.	0.803	0.172	0.497	—	—	—	—	—	0.591
Linear regression	15	0.230	—	−0.876	−0.237	—	—	—	—	—

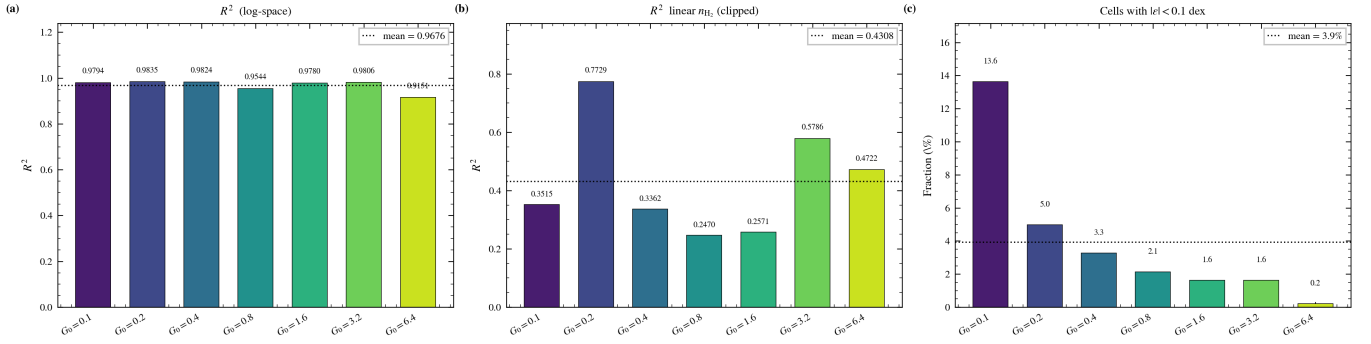


Figure 4. Three-panel summary of the final model (ENS_SP) performance. (a) Log-space R^2 per held-out fold. The two extrapolation folds ($G_0 = 0.1$ and $G_0 = 6.4$) are the most challenging; the dashed line marks the seven-fold mean. (b) Secondary clipped linear-space R^2 with predictions constrained to ± 1 dex of the true range before exponentiation; the substantially lower values illustrate the difficulty of linear-space density prediction over the full dynamic range. (c) Fraction of cells with absolute log-space residual below 0.1 dex. The concentration of accurate cells in mid-range folds indicates that the low- G_0 self-shielding-dominated regime is the most difficult to emulate accurately.

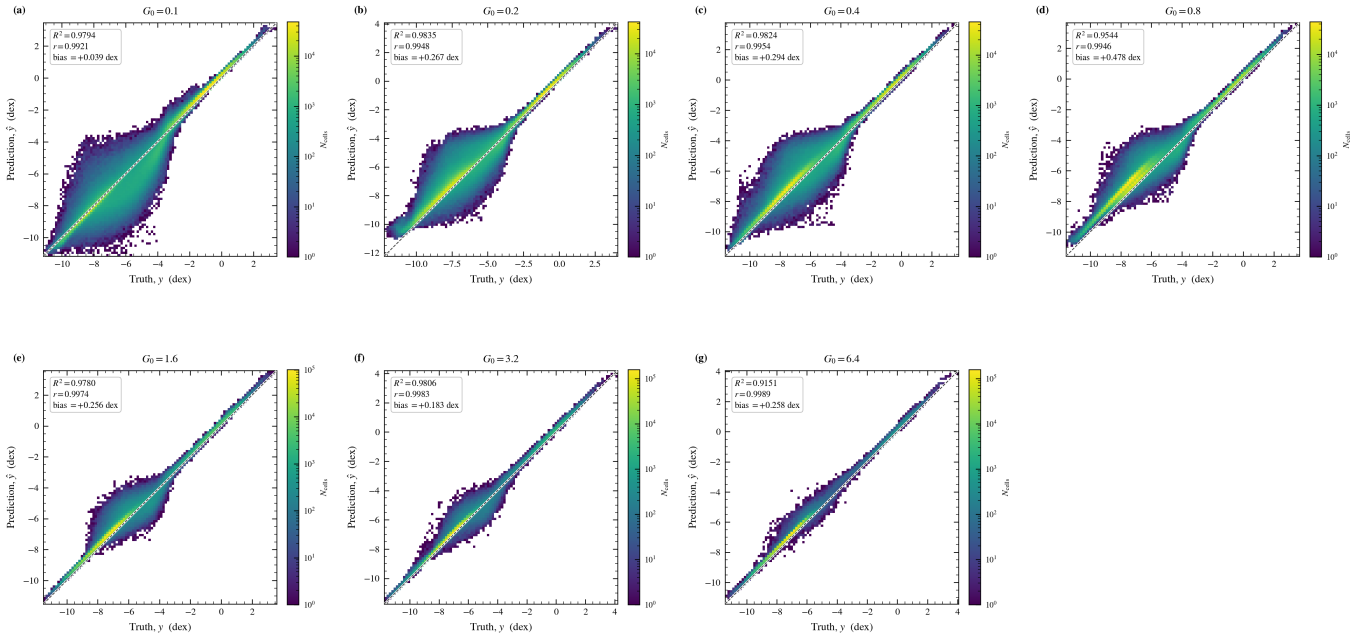


Figure 5. Predicted versus true $\log_{10}(n_{\text{H}_2})$ for all seven held-out folds. Each panel is a two-dimensional histogram with colour encoding point density on a logarithmic scale; the dashed line marks perfect prediction. Per-fold R^2 , RMSE (dex), and MAE (dex) are annotated in each panel. Scatter is widest at the two extrapolation folds ($G_0 = 0.1$ and $G_0 = 0.2$) and tightest at the mid-range interpolation folds ($G_0 = 0.8$ – 3.2).

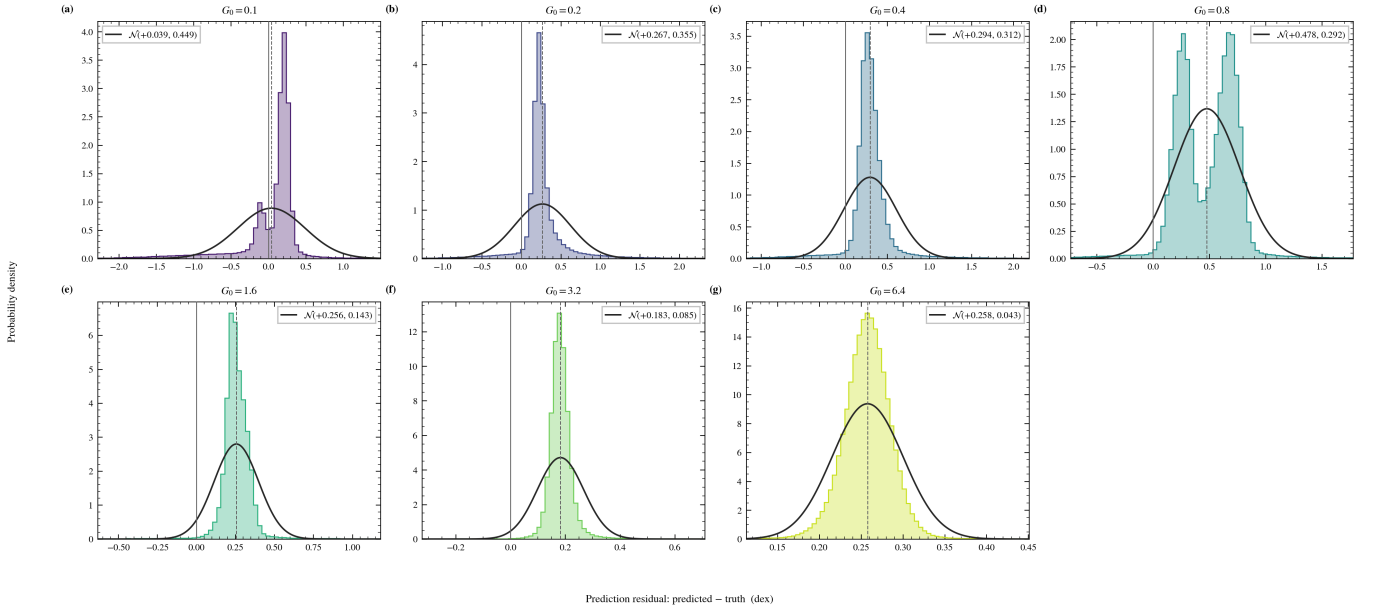


Figure 6. Residual distributions (prediction – truth, in dex) for all seven held-out folds. Histograms show the empirical distribution; curves show the best-fit normal $\mathcal{N}(\mu, \sigma)$ with parameters annotated. The distributions are approximately Gaussian and near-zero centred in all folds; the width decreases from the boundary folds ($G_0 = 0.1$ and $G_0 = 0.2$) to the interior folds ($G_0 = 1.6$ – 3.2).

5.3 Ensemble comparison

The equal-weight ensemble `ENS_SP` (mean $R^2 = 0.948$) marginally but consistently outperforms the Ridge meta-learner `STACKED_SP` (0.946). With only seven training folds, the meta-learner does not have sufficient meta-training signal to learn combination weights that transfer reliably to every held-out fold; equal averaging is the more robust choice in this low-fold-count regime. Including the 3D U-Net in any ensemble consistently *lowers* the mean R^2 : the `ENS_XGB+CNN` configuration achieves 0.897, worse than `XGB_STANDARD_SP` alone (0.924), because in the folds where the CNN predicts $R^2 < 0$, averaging with a well-calibrated tabular predictor degrades the result.

5.4 Spatial structure of predictions

Figure 9 shows representative 2D slices of the $\log_{10}(n_{H_2})$ field. At low G_0 the field displays rich filamentary spatial structure driven by self-shielding, with sharp transitions at cloud boundaries; at high G_0 the field is nearly uniform and residuals are minimal. This contrast explains the asymmetric difficulty of the boundary folds: the low- G_0 self-shielding morphology involves spatial gradients absent from the mid-range training cubes, while the high- G_0 UV-dominated morphology is a smooth continuation of the trend visible across the training range.

6 SUPPLEMENTARY EXPERIMENTS

6.1 Single-cube extrapolation: knowledge transfer across G_0

To quantify the information content of a single simulation and validate the cross-validation design, we train `STACKED_SP` on each individual cube and predict all seven, producing a 7×7 matrix of log-space R^2 values in which rows index training cubes and columns index test cubes.

Figure 10 shows these matrices for XGBoost, MLP, and the stacked

ensemble. The diagonal elements (in-sample fit) reach $R^2 \approx 0.99$. Off-diagonal elements decay monotonically with G_0 distance: one geometric step apart ($\times 2$ in G_0), $R^2 \approx 0.97$; two steps, ≈ 0.96 ; three steps, ≈ 0.88 ; four steps, ≈ 0.75 ; five steps, ≈ 0.45 ; six steps (i.e. $G_0 = 0.1$ to $G_0 = 6.4$), $R^2 \approx 0.15$. The mean off-diagonal R^2 across all 42 transfer pairs is approximately 0.78.

The local chemical equilibrium — the mapping from density, temperature, and shielding proxies to n_{H_2} — is approximately universal within a G_0 ratio of 4–8 but breaks down over larger factors. This is consistent with the physical picture that the H_2 formation and photodissociation rate functions are universal; what changes across G_0 is the equilibrium balance point. The result validates the leave-one- G_0 -out design by showing that a single-cube model is qualitatively weaker than the six-cube training ensemble, and that none of the 64-fold UV range is redundant in the training set.

6.2 Intra-cube spatial section: observational survey design

We ask whether the `ENS_SP` surrogate can reconstruct the n_{H_2} field within a single simulation from partial spatial coverage, and whether the *geometry* of the observed fraction matters more than its volume. For each of the seven G_0 cubes, we train `STACKED_SP` on a spatial subset of cells and evaluate on the complement, testing three sampling geometries at varying coverage fractions.

In the *random* configuration, cells are sampled uniformly at random, preserving the statistical distribution of local physical conditions across the training set. In the *contiguous slab* configuration, all cells from one spatial half of the cube along a coordinate axis are used for training. In the *contiguous box* configuration, a compact cubic sub-region of specified volume fraction is used.

Figure 11 shows the test R^2 as a function of geometry and coverage fraction for all seven G_0 cubes. The results are striking. A randomly sampled 1 per cent of cells ($\approx 21\,000$ cells) achieves test $R^2 = 0.897$ on the remaining 99 per cent. Increasing the fraction to 5, 10, 25, and 50 per cent yields R^2 of 0.952, 0.963, 0.970, and 0.980 respectively.

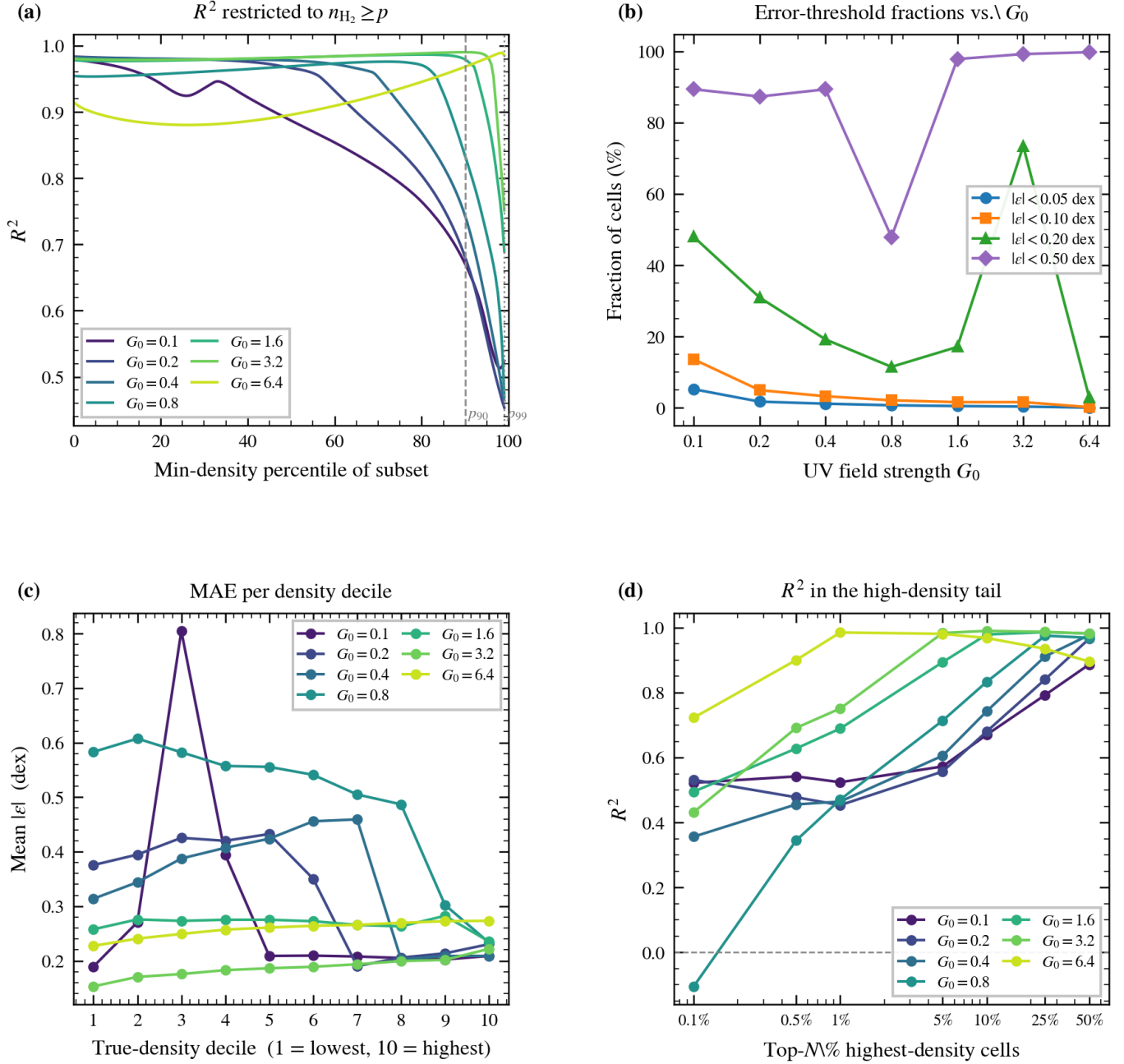


Figure 7. Density-stratified performance analysis for ENS_SP. (a) R^2 evaluated on cells above a minimum-density percentile; restricting to denser cells universally improves accuracy. (b) Fraction of cells with absolute residuals below thresholds of 0.05, 0.10, 0.20, and 0.50 dex as a function of G_0 . (c) MAE per density decile (decile 1 = lowest density); errors peak in decile 2, at the sharp H/H₂ phase transition. (d) R^2 within the top- M per cent highest-density cells; even the densest 1 per cent are predicted with $R^2 > 0.9$ for most folds.

By contrast, a contiguous slab covering 50 per cent of the volume — 23 times as many cells as the 1 per cent random sample — yields test $R^2 = -0.394$ (x-axis slab), -2.689 (y-axis), and -1.389 (z-axis).

The mechanism is the behaviour of the spatial neighbourhood features under different geometries. With random sampling, every test cell has training cells nearby in all three spatial dimensions; the box-filter neighbourhood features, computed from training cells, provide meaningful local averages for each test cell even at 1 per cent coverage. With contiguous-slab training, the model is trained exclusively on the UV-illuminated surface of the cloud and cannot

correctly predict the shielded interior: the feature distributions of interior cells fall entirely outside the training distribution, and the model defaults to an extrapolated mean that is systematically wrong. The z-axis slab is less catastrophic than the transverse slabs (z $R^2 = -1.4$ versus -2.7 for y) because the UV illumination direction along z produces a smoother gradient from surface to interior, making the half-cube feature distribution slightly more representative.

The practical consequence for observational astronomy is that an H₂ density field can be reconstructed from a sparse set of uniformly distributed sight-line observations — the kind obtainable from an

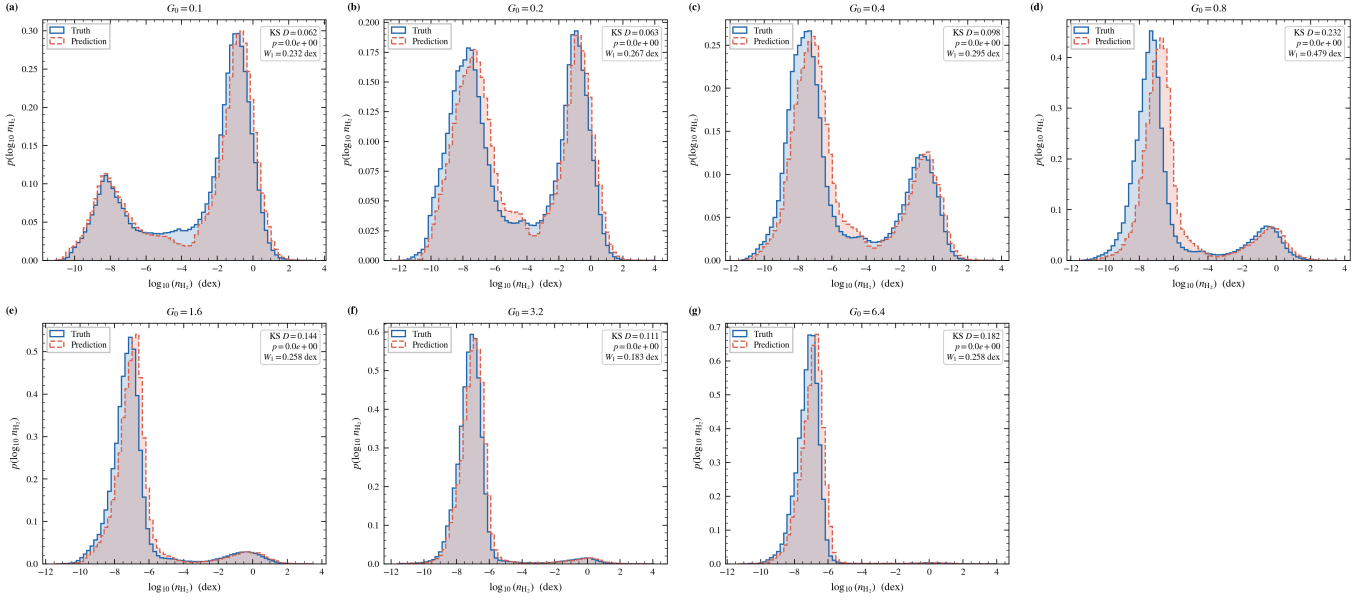


Figure 8. Marginal distributions of $\log_{10}(n_{H_2})$: true (blue, filled) and predicted (red, dashed outline) for each held-out fold. The ensemble accurately reproduces the bimodal structure at low G_0 and the narrower UV-dominated distribution at high G_0 . Per-fold R^2 , KL divergence, and MAE are annotated in each panel.

integral field unit (IFU) spectrograph with a sparse sampling grid — with $R^2 \approx 0.90$ at only 1 per cent spatial coverage. Deep, contiguous mapping of a single region consumes far more observing time while providing negligible reconstruction accuracy. Efficient H_2 density surveys should prioritise coverage uniformity over observational depth.

7 DISCUSSION

7.1 Feature engineering versus architecture search

The central methodological lesson of this work is that feature engineering dominates model architecture as a source of performance gain. The $+0.031$ R^2 improvement from the 3^3 spatial features is larger than any architecture change, and the full multi-scale extension to three kernel sizes ($+0.038$ combined) exceeds the ensemble gain ($+0.024$). This is consistent with the finding of Grinsztajn et al. (2022) that tree-based models with engineered features routinely outperform deep learning on tabular regression problems when training data volume is limited. The specific features chosen here — uniform box-filter means — are the simplest possible representation of spatial context; more sophisticated descriptors (local gradients, anisotropic kernels aligned with the UV illumination axis) could in principle improve accuracy further, particularly for the extrapolation folds.

7.2 Volumetric versus per-cell representations

The 3D U-Net has the nominally correct inductive bias for a spatially correlated field: its hierarchical receptive field should, in principle, learn to aggregate the shielding column density in a physically meaningful way. In practice, the mismatch between volumetric sample count (48 augmented cubes) and model parameters (5.8M for UNET_STANDARD) prevents the development of generalizable spatial representations, and the additional resolution loss from down-sampling ($128^3 \rightarrow 64^3$) discards the fine-scale gradients at cloud boundaries where n_{H_2} varies most sharply. This identifies a general

principle for simulation emulation with limited data: the choice between volumetric and per-cell approaches should be governed by the training sample-to-parameter ratio, not the nominal spatial nature of the problem. Volumetric models become competitive when dozens to hundreds of high-fidelity simulation volumes are available.

7.3 Regularisation in the limited-parameter regime

The catastrophic failure of dropout in the extrapolation folds reflects a tension between standard regularisation wisdom and the requirements of physical extrapolation. Dropout prevents overfitting by forcing redundant feature representations, which is beneficial when the model is memorising noise. In this problem, where the model must form coherent representations of a physical process over only six discrete G_0 training values, dropout prevents the *structure* of the learned representation from being preserved under extrapolation. This finding may be relevant to other simulation emulation contexts where training covers only a small number of discrete parameter values.

7.4 Limitations

Several limitations apply. The UV-only suite models chemistry under a spatially uniform, time-steady UV field; real molecular clouds experience anisotropic and time-varying radiation from nearby massive stars. The z -preserving augmentation strategy would require modification for simulations with different illumination geometries. The feature set includes $\log f_{H_2}$, a non-local quantity computed from the full 3D H_2 column density; in a deployment scenario where n_{H_2} is unknown, f_{H_2} would need to be estimated from other tracers, and that uncertainty would propagate into the n_{H_2} prediction. Finally, the data set represents a single turbulent ISM realisation at each G_0 value; generalisation across different initial conditions, magnetic field strengths, and density regimes remains to be quantified.

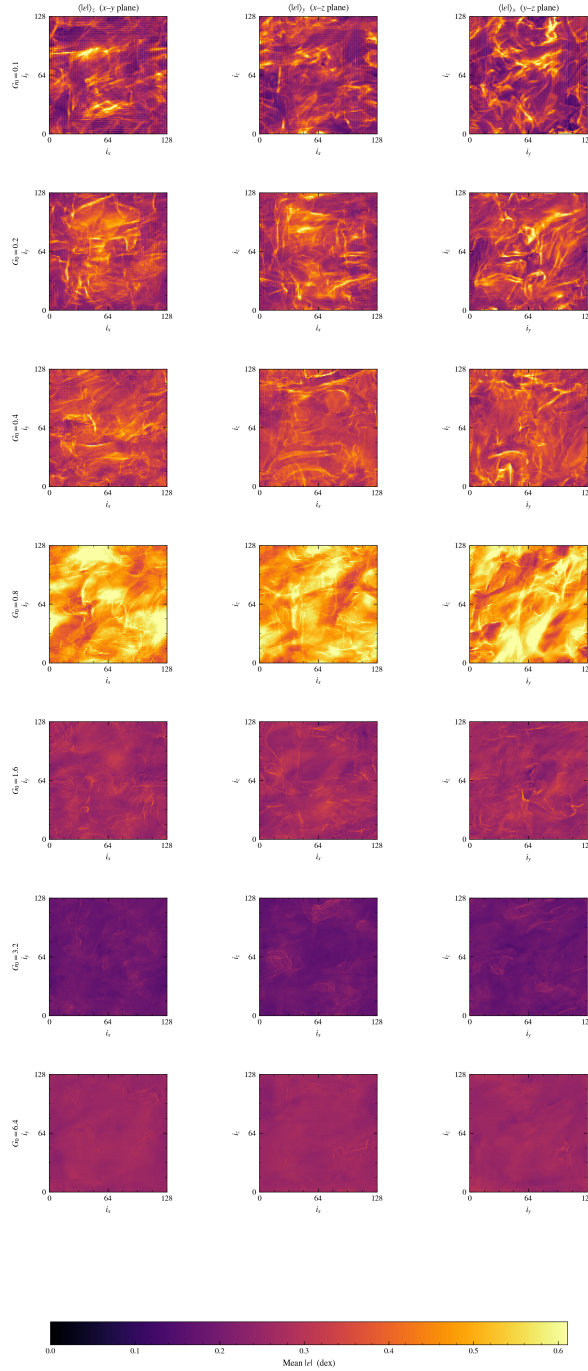


Figure 9. Representative 2D mid-plane slices of the $\log_{10}(n_{\text{H}_2})$ field across all seven G_0 values. Each row shows the ground truth, prediction, and absolute error (columns) for one fold. At low G_0 (top rows) the field has rich filamentary structure and residuals are elevated at cloud boundaries. At high G_0 (bottom rows) the field is nearly uniform and residuals are small. Colour encodes $\log_{10}(n_{\text{H}_2})$ on a common scale.

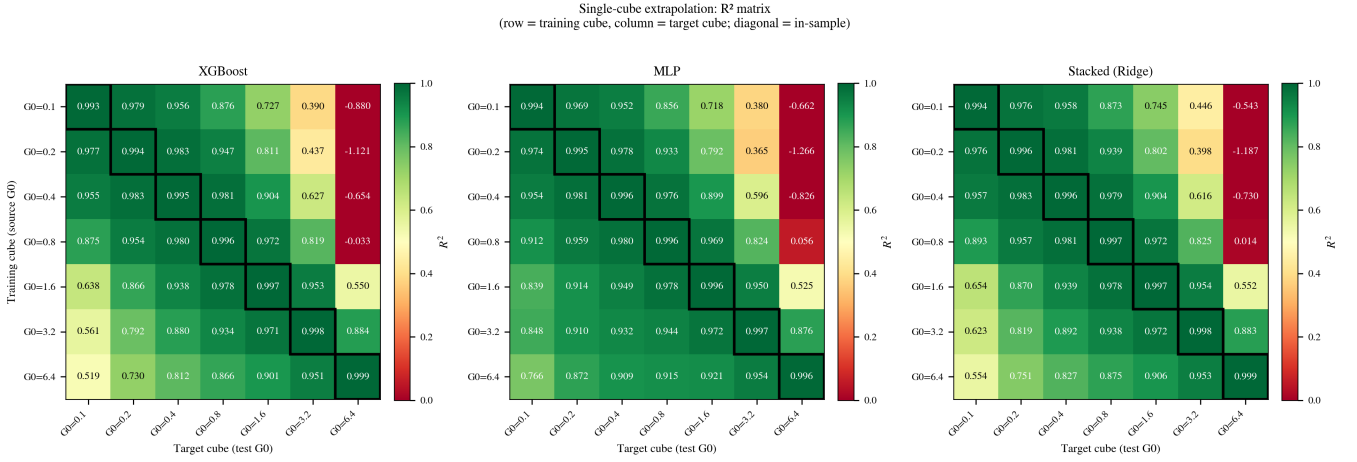


Figure 10. Single-cube extrapolation: 7×7 log-space R^2 matrices for XGBoost (left), MLP (centre), and stacked ensemble (right). Rows index training cubes; columns index test cubes; the diagonal (bold outline) gives in-sample fit ($R^2 \approx 0.99$). Colours range from red ($R^2 < 0$) through white (0.5) to green ($R^2 > 0.9$). Off-diagonal values decay monotonically with G_0 distance: chemical knowledge transfers well within $\times 4$ in G_0 but degrades rapidly at larger separations.

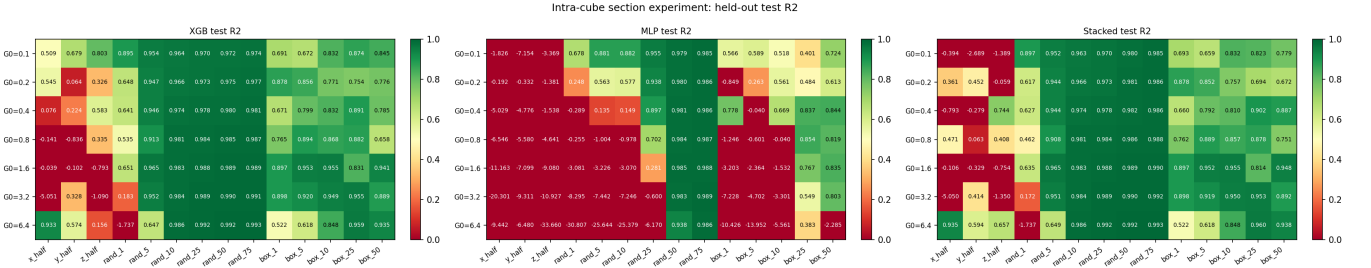


Figure 11. Intra-cube spatial section: test R^2 as a function of training geometry (columns) and source G_0 (rows) for XGBoost (left), MLP (centre), and stacked ensemble (right). Random sampling at 1 per cent achieves $R^2 \approx 0.90$ (green); contiguous slabs at 50 per cent coverage achieve $R^2 \ll 0$ (red), worse than predicting the training mean at every test cell. Contiguous boxes are intermediate. The pattern is consistent across all seven G_0 values.

8 CONCLUSIONS

We have developed and systematically evaluated ML surrogates for the molecular hydrogen number density in 3D ISM simulations, using leave-one- G_0 -out cross-validation to measure out-of-distribution generalisation across the dominant physical parameter. Our main conclusions are as follows.

(i) An equal-weight ensemble of XGBoost and a wide MLP, each trained on a 60-dimensional feature vector combining 15 local physical fields with 45 multi-scale spatial neighbourhood averages (kernel sizes 3^3 , 5^3 , 7^3), achieves a mean log-space R^2 of 0.948 ± 0.024 across seven leave-one- G_0 -out folds, with a minimum per-fold R^2 of 0.908 at the hardest extrapolation fold ($G_0 = 0.1$).

(ii) The single largest performance improvement (+0.031) comes from adding 3^3 spatial neighbourhood features, not from any model architecture change. Neural architecture search (three MLP designs, three XGBoost depths) yields differences smaller than 0.003.

(iii) A 3D U-Net CNN achieves a best mean R^2 of 0.803 ± 0.172 with an order of magnitude greater fold-to-fold variance, owing to the mismatch between volumetric training sample count (48 augmented cubes) and model parameters (5.8M). Including the CNN in any ensemble reduces the mean R^2 .

(iv) Dropout regularisation catastrophically degrades extrapolation to unseen G_0 values when training covers only six discrete UV

field strengths. L2 weight decay is the appropriate regulariser in this regime.

(v) Per-fold target normalisation (standardisation of $\log_{10}(n_{H_2})$ per training fold) is essential for CNN stability: without it, $R^2 < 0$; with it, $R^2 = 0.775$ immediately.

(vi) Single-cube extrapolation experiments confirm that chemical knowledge transfers well over G_0 ratios up to $\sim 4\text{--}8\times$ ($R^2 > 0.88$) but degrades rapidly over larger factors, validating the leave-one- G_0 -out protocol.

(vii) Intra-cube spatial sampling experiments demonstrate that 1 per cent uniform random coverage reconstructs the n_{H_2} field with $R^2 = 0.897$, while 50 per cent contiguous-slab coverage yields $R^2 = -0.394$. Coverage uniformity is the critical design parameter for observational H_2 density surveys; surveys targeting molecular cloud sight-lines should prioritise spatial uniformity over contiguous depth.

The ENS_SP surrogate predicts n_{H_2} at all 2,097,152 cells of a 128^3 cube in seconds on a single GPU, with accuracy within $\sim 0.1\text{--}0.2$ dex of the self-consistent simulation value across a 64-fold UV field range.

ACKNOWLEDGEMENTS

The author thanks the developers of XGBoost, PyTorch, NumPy, SciPy, and Matplotlib for the open-source tools on which this work depends.

DATA AVAILABILITY

All trained model outputs, prediction arrays, and experimental logs are archived as timestamped JSON and `.npz` files. The full pipeline is deterministic and runs end-to-end on a single consumer GPU.

REFERENCES

- Bisbas T. G., van Dishoeck E. F., Papadopoulos P. P., Szűcs L., Bialy S., Zhang Z.-Y., 2021, *ApJ*, 912, 105
- Cazaux S., Tielens A. G. G. M., 2004, *ApJ*, 604, 222
- Chen T., Guestrin C., 2016, in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, pp 785–794
- Draine B. T., Bertoldi F., 1996, *ApJ*, 468, 269
- Glover S. C. O., Mac Low M.-M., 2007, *ApJS*, 169, 239
- Glover S. C. O., Federrath C., Mac Low M.-M., Klessen R. S., 2010, *MNRAS*, 404, 2
- Grassi T., Bovino S., Schleicher D. R. G., Prieto J., Seifried D., Simoncini E., Gianturco F. A., 2014, *MNRAS*, 439, 2386
- Grinsztajn L., Oyallon E., Varoquaux G., 2022, in *Advances in Neural Information Processing Systems*.
- Habing H. J., 1968, *Bull. Astron. Inst. Netherlands*, 19, 421
- He K., Zhang X., Ren S., Sun J., 2016, in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*. pp 770–778
- Ioffe S., Szegedy C., 2015, *arXiv e-prints*, p. [arXiv:1502.03167](https://arxiv.org/abs/1502.03167)
- Katz H., et al., 2022, *MNRAS*, 510, 5603
- Kingma D. P., Ba J., 2015, in *Proc. 3rd International Conference on Learning Representations (ICLR)*.
- Narayanan D., Krumholz M. R., Ostriker E. C., Hernquist L., 2012, *MNRAS*, 421, 3127
- Ronneberger O., Fischer P., Brox T., 2015, in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, LNCS. Springer, pp 234–241
- Srivastava N., Hinton G., Krizhevsky A., Sutskever I., Salakhutdinov R., 2014, *J. Mach. Learn. Res.*, 15, 1929
- Ulyanov D., Vedaldi A., Lempitsky V., 2017, *arXiv e-prints*, p. [arXiv:1607.08022](https://arxiv.org/abs/1607.08022)
- Wolcott-Green J., Haiman Z., Bryan G. L., 2011, *MNRAS*, 418, 838